

ON PERMUTATION GROUPS AND THE O'NAN-SCOTT THEOREM

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Declaration

I, Rudrajit Dey, 7473U216015., hereby declare that this project entitled "**On Permutation Groups and the O’Nan Scott Theorem**" represents my original research/review work carried out as a B. Sc. (Hons.) student of Institute of Mathematics and Applications, Bhubaneswar and, to the best of my knowledge, it is not a complete copy of previously published or written by another person, nor any material presented for award of any other degree or diploma of IMA, Bhubaneswar or any other institution. Any contribution made to this research by others, with whom I have worked at IMA, Bhubaneswar or elsewhere, is explicitly acknowledged in the dissertation. Work of other authors cited in this dissertation have been duly acknowledged under the section Bibliography.

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Certificate

This is to certify that the work presented in this project entitled "**On Permutation Groups and the O’Nan-Scott Theorem**" by Rudrajit Dey, Roll no.: 7473U216015 is a record of original research/ review work carried out by him under my supervision and guidance in partial fulfillment of the requirements of the B. Sc. (Hons.) in Mathematics and Computing. Neither this project nor any part of it has been submitted for any degree or diploma to any institute or university in India or abroad.

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Abstract

Permutation groups arguably form the oldest part of group theory. Their study dates back to the early years of the nineteenth century and, indeed, for a long time groups were always understood to be permutation groups. Although, this is no longer true, permutation groups continue to play an important role in modern group theory through the ubiquity of group actions and the concrete representations which permutation groups provide for abstract groups. Today, both finite and infinite permutation groups are lively topics of research.

In this project we describe the development of permutation groups and the kinds of problems in finite permutation groups which can be usefully tackled using the classification of finite simple groups. We also state the different classification of the finite simple groups, i.e., the O’Nan-Scott Theorem, and the classifications are explained. We also state three applications of this theorem in a brief manner.

Keywords: Permutation Groups | Group Actions | Finite simple groups

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1 History and Literature

The study of groups originally grew out of an understanding of permutation groups. Permutations themselves had been studied extensively by Lagrange in 1770 in his work on the algebraic solutions of the polynomial equations. This subject flourished and by the mid-19th century a well-developed theory of permutation groups existed, codified by Camille Jordan in his book "*Traité des Substitutions et des Équations Algébriques*" of 1870. Jordan's book was in term, based on the papers that were left by Évariste Galois in 1832. When Cayley introduced the concept of an abstract group, it was not immediately clear whether or not this was a larger collection of objects than the known permutation groups. Cayley went on to prove that the concepts were equivalent in Cayley's theorem.

Our main literature that we follow is [1]. We study the properties and structures from this book along with the proof of the O'Nan Scott Theorem which was first announced by M. O'Nan and L.L.Scott at the Santa Cruz Conference on Finite Groups in 1979 (see [2]) and a complete proof appeared first in Buekenhout (1988). We then turn our focus on the Sim's Conjecture [3] which first appeared due to his investigation of the dihedral group D_{2p} having a representation as a primitive permutation group of degree p for cases of subdegree $d = 2$ and $d = 3$. It was finally proved by Peter Cameron [4].

2 Introduction

We begin with some important definitions and theorems related to permutation groups.

Definition 2.1. Let Ω be an arbitrary non-empty set, we refer to its elements as "points". A bijective mapping $\sigma : \Omega \rightarrow \Omega$ is called a "permutation" of Ω .

Proposition 2.1. The set of all permutation forms a group under composition of mappings, called the **Symmetric Group** denoted by $\text{Sym}(\Omega)$.

Proof. • If $\sigma_1, \sigma_2 \in \text{Sym}(\Omega) \implies \sigma_1 \circ \sigma_2 : \Omega \rightarrow \Omega$ is also bijective which implies $\sigma_1 \circ \sigma_2 \in \Omega$.

- If $\sigma_1, \sigma_2, \sigma_3 \in \text{Sym}(\Omega)$, then $\sigma_1 \circ (\sigma_2 \circ \sigma_3) = (\sigma_1 \circ \sigma_2) \circ \sigma_3$.
- We define $I(\alpha_1, \alpha_2, \dots) = (\alpha_1, \alpha_2, \dots)$ where $\alpha_i \in \Omega$. Then, I is the identity of $\text{Sym}(\Omega)$, since

$$\sigma \circ I(\alpha_1, \alpha_2, \dots) = \sigma(\alpha_1, \alpha_2, \dots) = \sigma$$

$$I \circ \sigma(\alpha_1, \alpha_2, \dots) = \sigma(\alpha_1, \alpha_2, \dots) = \sigma$$

- Since σ is bijective $\implies \sigma^{-1}$ exists $\sigma \circ \sigma^{-1} = I = \sigma^{-1} \circ \sigma$

□

Note : S_n is used to denote a special symmetric group of n positive integers $\Omega = \{1, 2, \dots, n\}$

Definition 2.2. A permutation group is a subgroup of symmetric group.

If Ω and Ω' are two non-empty sets of the same cardinality (i.e. $\exists$ a bijection $\alpha \mapsto \alpha'$ from $\Omega \mapsto \Omega'$) then the group $\text{Sym}(\Omega)$ is isomorphic to the group $\text{Sym}(\Omega')$ via the mapping $\sigma \mapsto \sigma'$ defined by : $\sigma'(\alpha') = \beta'$ whenever $\sigma(\alpha) = \beta$. In particular, $\text{Sym}(\Omega') \cong \text{Sym}(\Omega)$.

There are two common ways in which permutations are represented : (for the finite case)

1. The mapping $\sigma : \Omega \rightarrow \Omega'$ can be written explicitly in the form: $\sigma = \begin{pmatrix} \alpha_1 & \alpha_2 & \dots & \alpha_n \\ \beta_1 & \beta_2 & \dots & \beta_n \end{pmatrix}$ where the first row is some enumeration of points of Ω and β_i is the image of α_i under σ .

2. Another notation is to write σ as a product of disjoint cycles. A permutation ρ is called an r -cycle ($r \in \mathbb{N}$) if for r distinct points $a_1, a_2, \dots, a_r \in \Omega$, ρ maps $a_i \mapsto a_{i+1}$ ($i = 1, \dots, r-1$), maps $a_r \mapsto a_1$ and leaves all other points as fixed. ρ is an infinite cycle if for some doubly infinite sequence $a_i, i \in \mathbb{Z}$, ρ maps a_i onto $a_{i+1} \forall i$ and leaves other points fixed. By "disjoint" we mean that no two cycles move a common point.

Example: Let $\Omega = \{0, 1, \dots, 6\}$ be the finite field of 7 elements under addition and multiplication modulo 7. Then, the mapping $\alpha \mapsto 4\alpha + 1$ defines a permutation of Ω . This permutation can be denoted as: $\begin{pmatrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 5 & 2 & 6 & 3 & 0 & 4 \end{pmatrix}$ or $(015)(2)(364) = (015)(364)$

Proposition 2.2. An r -cycle $(\alpha_1 \dots \alpha_r)$ is equal to an s -cycle $(\beta_1 \dots \beta_s)$ on the same set Ω iff $r = s$ and for some h we have $\alpha_{i+h} = \beta_i$ for each i where the indices are taken modulo r . Also two infinite cycles $(\dots \alpha_{-1} \alpha_0 \alpha_1 \dots)$ and $(\dots \beta_{-1} \beta_0 \beta_1 \dots)$ on the same set are equal iff for some h , $\alpha_{i+h} = \beta_i \forall i$

Proof. $\implies$ Given that an r -cycle is equal to an s -cycle, i.e., $(\alpha_1 \dots \alpha_r) = (\beta_1 \dots \beta_s)$ on the same set Ω , it is easy to observe that the length of the two cycles are equal i.e., $r = s$ and both the cycles permute the same elements, thus, $(\beta_1 \dots \beta_r) = (\alpha_i \alpha_{i+1} \dots \alpha_j)$ where $(i, \dots, j) = (1, \dots, r) \implies \alpha_{i+h} = \beta_i (i+h) \text{ mod } r$.

($\Leftarrow$) Now, given that for an r -cycle and s -cycle $r = s$ and $\alpha_{i+h} = \beta_i (i+h) \text{ mod } r$. It is easily seen that both the cycles represent the same permutation upto ordering of the elements of the cycle.

For 2 infinite cycles $(\dots \alpha_{-1} \alpha_0 \alpha_1 \dots)$ and $(\dots \beta_{-1} \beta_0 \beta_1 \dots)$ to be equal, they must represent the cycle of same elements from the same set. Hence, the elements are equal upto cyclic permutation, i.e., $\alpha_{i+h} = \beta_i \forall i$ $\square$

Proposition 2.3. Let G be a group acting on the non-empty set A . The relation on A defined by $a \sim b$ iff $a = b^g = g \cdot b$ for some $g \in G$ is an equivalence relation. For each $a \in A$, the no. of elements in the equivalence class containing A is the index of the stabilizer of a : $|G : G_a|$.

Proof. We first show that $\sim$ is an equivalence relation:

- $a \sim a$ since $a = e \cdot a, e \in G$ (reflexive).
- $a \sim b \implies a = g \cdot b \implies g^{-1} \cdot a = b \implies b \sim a$ (symmetric)
- $a \sim b$ and $b \sim c \implies a = g_1 \cdot b$ and $b = g_2 \cdot c \implies a = g_1 \cdot (g_2 \cdot c) = (g_1 g_2) \cdot c \implies a \sim c$ (transitive)

Thus, $\sim$ partitions A into equivalence classes of each $a \in A$. Consider, $C_a = \{g \cdot a | g \in G\}$ be the equivalence class of a . Suppose, $b = g \cdot a \in C_a$. Then, gG_a is a left coset of G_a in G . Consider the map $b = g \cdot a \mapsto gG_a$ from C_a to the left cosets of G_a in G . This map is surjective, since for any $g \in G$, the element $g \cdot a \in C_a$. Also, $g \cdot a = h \cdot a$ iff $h^{-1}g \in G_a$ iff $gG_a = hG_a$, thus, this map is also injective. $\square$

Theorem 2.1. *Each permutation $\sigma \in \text{Sym}(\Omega)$ can be written as a product of disjoint cycles. This product is unique upto ordering in which the cycles appear in the product and the inclusion-exclusion of 1-cycles*

Proof. Suppose Ω is a finite set, then, $\text{Sym}(\Omega)$ is finite. We show that every element of the symmetric group $\text{Sym}(\Omega)$ has a unique cycle decomposition. Let $\Omega = \{a_1, a_2, \dots, a_n\}$, $\sigma \in \text{Sym}(\Omega)$ and $G = \langle \sigma \rangle$.

Thus, G acts on Ω and so by 2.3, it partitions $\{a_1, a_2, \dots, a_n\}$ into unique set of (disjoint) orbits. Let O be one of these orbits and let $x \in O$. By 2.3 applied to $\Omega = O$, we see there is a bijection between the left cosets of G_x in G and the elements of O , given explicitly by $\sigma^i x \mapsto \sigma^i G_x$. Since, G is a cyclic group, $G_x \trianglelefteq G$ and G/G_x is cyclic of order d , where d is the smallest positive integer for which $\sigma^d \in G_x$. Also, $d = |G : G_x| = |O|$.

Thus, the distinct cosets of G_x in G are: $1G_x, \sigma G_x, \sigma^2 G_x, \dots, \sigma^{d-1} G_x$. This shows that the distinct elements of O are: $x, \sigma(x), \sigma^2(x), \dots, \sigma^{d-1}(x)$. Ordering the elements of O in this manner shows that σ cycles the elements of O , i.e., on an orbit of size d , σ acts as a d -cycle. This proves the existence of a cyclic decomposition for each $\sigma \in \text{Sym}(\Omega)$. The orbits of $\langle \sigma \rangle$ are uniquely determined by σ .

The only latitude is in which order the orbits are listed. Within each orbit O , we may begin with any element as a representative. Choosing $\sigma^i(x)$ instead of x as the initial representative simply produces the elements of O in the order: $\sigma^i(x), \sigma^{i+1}(x), \dots, \sigma^{d-1}(x), x, \sigma(x), \dots, \sigma^{i-1}(x)$ which is the cyclic permutation of the original list. It follows that the decomposition is unique upto rearrangement of the cycles and upto a cyclic permutation of the integers within the cycle. $\square$

2.1 Group Actions

Definition 2.3. *Let G be a group and Ω be a non-empty set. Suppose for each $\alpha \in \Omega$ and each $x \in G$ we have defined an element of Ω denoted by α^x (in other words, $(\alpha, x) \mapsto \alpha^x$ as a function of $\Omega \times G$ into Ω), then we say that defines an **action** of G on Ω if*

1. $\alpha^1 = \alpha$
2. $(\alpha^x)^y = \alpha^{xy} \quad \forall \alpha \in \Omega \text{ and } x, y \in G.$

Every subgroup G of $\text{Sym}(\Omega)$ acts naturally on Ω where α^x is simply the image of α under permutation x . If a group G acts on a non-empty set Ω , then to each element $x \in G$, we can associate a mapping $\bar{x}$ of Ω into itself, namely, $\alpha \mapsto \alpha^x$. The mapping $\bar{x}$ itself is a bijection since it has $\bar{x}^{-1}$ as its inverse; hence, we have a mapping $\rho : G \mapsto \text{Sym}(\Omega)$ given by $\rho(x) = \bar{x}$. Moreover, using the definition of group actions, we see that ρ is a group homomorphism since $\forall \alpha \in \Omega$ and $x, y \in G$, the image of α under $\overline{xy}$ is same as its image under the product $\bar{x} \cdot \bar{y}$. In general, any homomorphism of G into $\text{Sym}(\Omega)$ is called a *representation* of G on Ω . Hence, we see that each action of G on Ω gives rise to a representation of G on Ω .

Definition 2.4. *We provide definitions of some well known terms related to group actions:*

- The **degree** of an action is the size of Ω .
- The **kernel** of an action is the kernel of representation of G on Ω .
- An action is **faithful** when $\ker \rho = 1$.

Definition 2.5. (Cayley Representation): For any group G , we can take $\Omega := G$ and define an action by right multiplication as follows : a^x with $a, ax \in \Omega$ and $x \in G$. The corresponding representation of G into $\text{Sym}(\Omega)$ is called **regular representation**. It is faithful since the kernel, $\ker \rho = \{x \in G | a^x = a \forall a \in \Omega\} = 1$.

This shows that every group is isomorphic to a permutation group.

Action on Right Cosets: For any group G and any subgroup H of G , we can consider $\Gamma_H = \{H \cdot a | a \in G\}$ as the set of right cosets of H in G , and define an action of G on Γ_H by right product: $(H \cdot a)^x := Hax$, where $Ha, Hax \in \Gamma_H$ and $x \in G$. We denote the corresponding representation of G on Γ_H by ρ_H . Since $Hax = Ha \Leftrightarrow x \in a^{-1}Ha$ we have, $\ker \rho_H = \bigcap_{a \in G} a^{-1}Ha$, in general, ρ_H is not faithful.

Definition 2.6. We now provide definitions for two important structures that come up when a group acts on a set.

- When a group G acts on a set Ω , a typical point α is moved by elements of G to various other points. The set of these images is called the "**orbit**" of α under G , denoted by : $\alpha^G := \{\alpha^x | x \in G\}$.
- The set of elements in G which fix a specific point $\alpha \in \Omega$ acts on Ω is called the "**stabilizer**" of α in G denoted by : $G_\alpha := \{x \in G | \alpha^x = \alpha\}$.

The important properties of these objects are summarized in the following theorem:

Theorem 2.2. Suppose that G is a group acting on a set Ω and $x, y \in G$ and $\alpha, \beta \in \Omega$. Then,

- Two orbits α^G and β^G are either equal or disjoint, so the set of all orbits is a partition of Ω into mutually disjoint subsets.
- The stabilizer G_α is a subgroup of G and $G_\beta = x^{-1} \cdot G_\alpha \cdot x$ whenever $\beta^x = \alpha$. Moreover, $\alpha^x = \alpha^y \Leftrightarrow G_\alpha x = G_\alpha y$
- (Orbit-Stablizer Property) : $|\alpha^G| = |G : G_\alpha| \quad \forall \alpha \in \Omega$.

Proof. • If $\delta \in \alpha^G$ then $\delta = \alpha^u$ for some $u \in G$. Since ux runs over the elements of G as x runs over G , $\delta^G = \{\delta^x | x \in G\} = \{\alpha^{ux} | x \in G\} = \alpha^G$. Hence, if $\delta \in \beta^G$ as well, then, $\Rightarrow \delta^G = \beta^G \Rightarrow \alpha^G = \delta^G = \beta^G$. Since every element $\alpha \in \Omega$ lies in atleast one orbit (equivalence classes, namely α^G), thus proved.

- Clearly, $1 \in G_\alpha$, and whenever $x, y \in G_\alpha$ then $xy^{-1} \in G_\alpha$. Thus, G_α is a subgroup of G . If $\beta = \alpha^x$, then we also have : $y \in G_\beta \Leftrightarrow \alpha^{xy} = \alpha^x \Leftrightarrow xyx^{-1} \in G_\alpha$ and hence $x^{-1}G_\alpha x = G_\beta$. Also, $\alpha^x = \alpha^y \Leftrightarrow \alpha^{xy^{-1}} = \alpha \Leftrightarrow xy^{-1} \in G_\alpha \Leftrightarrow G_\alpha x = G_\alpha y$.
- This follows immediately since the second statement shows that the distinct points in α_G are in bijective correspondence with the right cosets of G_α in G , thus, $|\alpha^G| = |G : G_\alpha|$ which for the finite case is equal to $\frac{|G|}{|G_\alpha|}$.

□

A group G acting on a set Ω is said to be *transitive* on Ω if it has only one orbit, and so $\alpha^G = \Omega$ for all $\alpha \in \Omega$. Equivalently, G is transitive if for every pair of points $\alpha, \beta \in \Omega$ there exists $x \in G$ such that $\alpha^x = \beta$. A group which is not transitive is called *intransitive*. A group G acting transitively on a set Ω is said to act *regularly* if $G_\alpha = 1$ for each $\alpha \in \Omega$ (equivalently, only the identity fixes any point).

2.2 Blocks and Primitivity

Definition 2.7. Let G be a group acting transitively on a set Ω . A non-empty subset Δ of Ω is called a *block* of G if for each $x \in G$ either $\Delta^x = \Delta$ or $\Delta^x \cap \Delta = \emptyset$ where $\Delta^x := \{\gamma^x | \gamma \in \Delta\}$.

Every group G acting transitively on Ω has Ω and the singleton sets $\{\alpha\}$ as blocks, called the **trivial blocks**. A block which is minimal in the set of all blocks is called the **minimal block** (of size > 1). If G acts transitively on Ω and Δ, Γ are blocks for G , then $\Delta \cap \Gamma$ is also a block for G . Generally, any intersection of blocks containing a common point is a block for G .

We now show an important structure based upon blocks called the "**system of blocks**".

Proposition 2.4. If G is a group acting on a set Ω , then a G -congruence on Ω is an equivalence relation on Ω denoted by $\approx$, with the property that $\alpha \approx \beta \Leftrightarrow \alpha^x \approx \beta^x \forall x \in G$. If G acts transitively on Ω and $\approx$ is a G -congruence, then the equivalence classes of $\approx$ form a system of blocks for G .

Proof. Since G acts transitively on Ω this implies that for every pair of points $\alpha, \beta \in \Omega, \exists x \in G$ such that $\alpha^x = \beta$. Let $[\alpha]$ denote the equivalence class of α i.e., $[\alpha] = \{\beta \in \Omega | \alpha \approx \beta\}$.

Consider any point $b \in [\alpha]$. Then we have, $\alpha \approx \beta \implies \alpha^x \approx \beta^x$, also, $\alpha^x = b \implies b \approx b^x \implies b \in [b^x] = [\alpha^x] \implies [\alpha] \subseteq [\alpha^x] \dots (1)$

Also, $b^x \in [\alpha^x] \implies b^x \in [\alpha] (\because \text{transitivity of } \approx) \implies [\alpha]^x \subseteq \alpha \dots - (2).$

From (1) and (2) we can conclude : $[\alpha]^x = [\alpha]$.

Now, if $\alpha \not\approx \beta$, then we have, $[\alpha]^x \cap [\beta]^x = \{a^x | \alpha \approx a\} \cap \{b^x | \beta \approx b\} = \emptyset (\because \text{if } a^x \in [\alpha]^x \implies a^x \approx \alpha^x \implies a \approx \alpha \not\approx \beta \implies a \not\approx \beta \implies a \not\approx b \implies a \not\approx b^x \implies a \notin [\beta]^x).$

Hence, $\Sigma = \{[\alpha]^x | x \in G\}$ forms a system of blocks for G and the equivalence classes forms a partition of Ω . $\square$

Theorem 2.3. Suppose G acts transitively on Ω and Δ is a block of G . Put $\Sigma := \{\Delta^x | x \in G\}$. Then, the sets in Σ form a partition of Ω and each element of Σ is a block of G .

Proof. Since G acts transitively on Ω and Δ is a block of $G \implies \Delta^x = \Delta$ or $\Delta^x \cap \Delta = \emptyset$ for each $x \in G$. We have $\Sigma = \{\Delta^x | x \in G\}$. First we show that each element of Σ is a block for G .

Let $\Delta^x \in \Sigma$ for some fixed $x \in G$. We want to show for every $g \in G$, either $(\Delta^x)^g = \Delta^x$ or $(\Delta^x)^g \cap \Delta^x = \emptyset$.

Suppose $(\Delta^x)^g \cap \Delta^x \neq \emptyset$. Let $a \in (\Delta^x)^g \cap \Delta^x \implies \exists b \in \Delta$ such that $a = b^x (\because a \in \Delta^x)$ and $\exists c \in \Delta^x$ such that $a = c^g (\because a \in (\Delta^x)^g)$. Since, $c \in \Delta^x \implies \exists b' \in \Delta$ such that $c = (b')^x$ so that, $a = c^g = b'^{xg} = b'^{xg}$.

Now, consider $a^{x^{-1}}$. We have, $a^{x^{-1}} = (b^x)^{x^{-1}} = b^{xx^{-1}} = b^1 = b \in \Delta$ and $a^{x^{-1}} = b'^{xgx^{-1}} \in \Delta^{xgx^{-1}}$. So, $a^{x^{-1}} \in \Delta \cap \Delta^{xgx^{-1}} \implies \Delta \cap \Delta^{xgx^{-1}} \neq \emptyset$. Hence, $\Delta = \Delta^{xgx^{-1}} \implies \Delta^x = (\Delta^x)^g (\because \Delta^{xgx^{-1}}$ is also a block). Hence, every element of Σ is a block for G .

Now, Σ forms a partition of Ω by the above proposition 2.4. Hence, proved. $\square$

Definition 2.8. Suppose G is a group acting on a set $\Omega, \Delta \subseteq \Omega$. Then,

- The set $G_{(\Delta)} := \{x \in G \mid \delta^x = \delta, \forall \delta \in \Delta\}$ is called the **pointwise stabiliser** of Δ in G .
- The set $G_{\{\Delta\}} := \{x \in G \mid \Delta^x = \Delta\}$ is called the **setwise stabiliser** of Δ in G .

We can easily verify that $G_{(\Delta)}$ and $G_{\{\Delta\}}$ are both subgroups of G , and moreover $G_{(\Delta)} \triangleleft G_{\{\Delta\}}$. Note that, $G_{(\alpha)} = G_{\{\alpha\}} = G_\alpha, \forall \alpha \in \Omega$.

Remark: If G acts transitively on Ω and Δ is a block for G , then $G_{\{\Delta\}}$ acts transitively on Δ .

Proof. If G acts transitively on $\Omega \implies$ for each pair $\alpha, \beta \in \Omega, \exists x \in G$ such that $\alpha^x = \beta$. Given, Δ is a block for $G \implies \Delta^x = \Delta$ or $\Delta^x \cap \Delta = \emptyset, \forall x \in G$.

We want to show that $\alpha^{G_{\{\Delta\}}} = \Delta$, for $\alpha \in \Delta, x \in G_{\{\Delta\}}$ we have $G_{\{\Delta\}} = \{x \in G \mid \Delta^x = \Delta\}$.

Now, let $\alpha, \beta \in \Delta$, then $\exists x \in G_{\{\Delta\}}$ such that $\alpha^x = \beta$. Since, α, β are arbitrary, hence, $G_{\{\Delta\}}$ acts transitively on Δ . $\square$

Theorem 2.4. Let G be a group which acts transitively on a set Ω , and let $\alpha \in \Omega$. Let $\mathcal{B}$ be the set of all blocks Δ for G with $\alpha \in \Omega$, and let $\mathcal{S}$ denote the set of all subgroups H of G with $G_\alpha \leq H$. Then there is a bijection Ψ of $\mathcal{B}$ onto $\mathcal{S}$ given by $\Psi(\Delta) := G_{\{\Delta\}}$ whose inverse mapping Φ is given by $\Phi(H) := \alpha^H$. The mapping Ψ is order-preserving in the sense that if $\Delta, \Gamma \in \mathcal{B}$ then $\Delta \subseteq \Gamma \Leftrightarrow \Psi(\Delta) \leq \Psi(\Gamma)$.

Proof. We first show that Ψ maps $\mathcal{B}$ into $\mathcal{S}$. let $\Delta \in \mathcal{B}$. Then $x \in G_\alpha$ implies that $\alpha \in \Delta \cap \Delta^x$, and so $\Delta = \Delta^x$ because Δ is a block for G . This show that each $x \in G_\alpha$ lies in $G_{\{\Delta\}}$. Hence, $G_{\{\Delta\}} \supseteq G_\alpha$ for all $\Delta \in \mathcal{B}$ and so Ψ maps $\mathcal{B}$ into $\mathcal{S}$.

We next show that Φ maps $\mathcal{S}$ into $\mathcal{B}$. Let H be a subgroup of G with $G_\alpha \leq H$. Put $\Delta := \alpha^H$, and let $x \in G$. Clearly, $\Delta^x = \Delta$ if $x \in H$, and we claim that $\Delta^x \cap \Delta = \emptyset$ otherwise. Indeed if $\Delta^x \cap \Delta \neq \emptyset$, then there exists $u, v \in H$ such that $\alpha^{ux} = \alpha^v$. Then $uxv^{-1} \in G_\alpha$, and so $x \in u^{-1}G_\alpha v \subseteq H$. Thus, $\Delta^x \cap \Delta = \emptyset$ whenever $x \notin H$, and so Δ is a block which contains α , therefore lies in $\mathcal{B}$. Thus, Φ maps $\mathcal{S}$ into $\mathcal{B}$. Moreover, since Δ is an orbit for $G_{\{\Delta\}}$, the composite mapping of Ψ followed by Φ is the identity on $\mathcal{B}$.

To prove that Φ and Ψ are inverses, it remains to show that the composite of Φ followed by Ψ is the identity on $\mathcal{S}$. Let $H \in \mathcal{S}$, and put $\Delta := \Phi(H) = \alpha^H$. The previous paragraph shows that if $x \in G$, then $\Delta^x = \Delta \Leftrightarrow x \in H$. Thus, $H = G_{\{\Delta\}}$ as required. This completes the proof that Φ is the inverse of Ψ .

The statement that Ψ is order-preserving follows at once. Indeed $G_{\{\Delta\}} \subseteq G_{\{\Gamma\}}$ implies that the orbits of α under these groups (namely, Δ and Γ) satisfy $\Delta \subseteq \Gamma$. Conversely, if $\Delta \subseteq \Gamma$, then $x \in G_{\{\Delta\}}$ implies that $\Gamma^x \cap \Gamma \neq \emptyset$ and hence $x \in G_{\{\Gamma\}}$ because Γ is a block. Thus, $\Delta \subseteq \Gamma$ implies that $G_{\{\Delta\}} \leq G_{\{\Gamma\}}$. This shows that Ψ is order-preserving, and the theorem is proved. $\square$

2.3 Permutation Representations and Normal Subgroups

Let G be a group acting on a set Ω . A subset Γ of Ω is called *invariant* or *G -invariant* if $\Gamma^x = \Gamma$ for all $x \in G$. Clearly Γ is *G -invariant* $\Leftrightarrow \Gamma$ is union of orbits of G . Thus, if Γ is *G -invariant* then we can consider a restriction of the action of G to Γ and obtain an action of G on $\Gamma, x \mapsto x^\Gamma$ is the representation corresponding to this action on $\Gamma (x^\Gamma \in \text{Sym}(\Omega))$ is the permutation of Γ associated with $x \in G$) and we write $G^\Gamma := \{x^\Gamma \mid x \in G\}$. The representation $x \mapsto x^\Gamma$ is a homomorphism of G onto G^Γ with kernel $G_{(\Gamma)}$ and so by first isomorphism theorem : $\frac{G}{G_{(\Gamma)}} \cong G^\Gamma$.

Two permutation groups, say $G \leq \text{Sym}(\Omega)$ and $H \leq \text{Sym}(\Omega')$ are called **permutation isomorphic** if there exists a bijection $\lambda : \Omega \rightarrow \Omega'$ and a group isomorphism $\psi : G \rightarrow H$ such that

$$\lambda(\alpha^x) = \lambda(\alpha)^{\psi(x)} \text{ for all } \alpha \in \Omega \text{ and } x \in G.$$

Suppose that G is a group acting imprimitively on a set Ω , that H is a normal subgroup of G and that Σ is a system of blocks for G . If $\Delta, \Delta' \in \Sigma$, then $H^\Delta \leq \text{Sym}(\Delta)$ and $H^{\Delta'} \leq \text{Sym}(\Delta')$ are permutation isomorphic. Indeed, since Σ is a system of blocks we know that $\Delta = \Delta^c$ for some $c \in G$, and then we can define a bijection λ of Δ onto Δ' by $\lambda(\delta) := \delta^c$. Now we claim that we can define an isomorphism $\psi : H^\Delta \rightarrow H^{\Delta'}$ by $\psi(x^\Delta) := (c^{-1}xc)^{\Delta'}$. First, ψ is well-defined and injective since for all $x, y \in H$ we have $x^\Delta = y^\Delta \Leftrightarrow xy^{-1} \in H_{(\Delta)} \Leftrightarrow c^{-1}(xy^{-1})c \in H_{(\Delta')} \Leftrightarrow (c^{-1}xc)^{\Delta'} = (c^{-1}yc)^{\Delta'}$ because $\Delta' = \Delta^c$. Second, ψ is surjective since $c^{-1}Hc = H$. Finally, since $\psi(x^\Delta y^\Delta) = \psi((xy)^\Delta) = (c^{-1}(xy)c)^{\Delta'} = (c^{-1}xc)^{\Delta'}(c^{-1}yc)^{\Delta'} = \psi(x^\Delta)\psi(y^\Delta)$ for all $x, y \in H$, we conclude that ψ is an isomorphism as claimed. It is thus easy to see that λ and ψ define the required permutation isomorphism.

Suppose that the group G acts on a set Ω and let T be a subset of G . Then we define the *support* and set of *fixed-points* of T by:

$$\text{supp}(T) := \{\alpha \in \Omega \mid \alpha^x \neq \alpha \text{ for at least one } x \in T\}$$

and

$$\text{fix}(T) := \{\alpha \in \Omega \mid \alpha^x = \alpha \text{ for all } x \in T\}.$$

Note that Ω is the disjoint union of these two sets.

3 The O’Nan-Scott Theorem

The classification of finite simple groups has led to the solution of many problems in the theory of finite permutation groups. An important starting point in such applications is the reduction of the O’Nan-Scott Theorem ([2]) for primitive permutation groups. Before we state the theorem we would like to state some important definitions and properties of some structures that arise from the permutation groups.

3.1 Group Extensions

Definition 3.1. *If Q and N are two groups, then G is a **group extension** of Q by N if there exists a short exact sequence:*

$$1 \rightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} Q \rightarrow 1$$

such that $\iota : N \hookrightarrow G$ is injective of G and $\pi : G \twoheadrightarrow Q$ is surjective.

A group extension is a general means of describing a group in terms of a particular normal subgroup and quotient group. If G is an extension of Q by N , then $\iota(N)$ is a normal subgroup of G and the quotient group $\frac{G}{\iota(N)} \cong Q$. Since, any finite group has a *maximal normal subgroup* N , with *simple factor group* $\frac{G}{N}$, all finite groups may be constructed as a series of extensions with finite simple groups.

Given groups Q and N , there are (often) many extensions of Q by N . Examples include the direct product of Q and N and a semidirect product of Q and N . A function $\tau : Q \rightarrow G$ such that $\pi\tau$ is the identity function on Q , is called the *transversal function*. A group extension is said to be **split** if there is a transversal function which is a homomorphism.

3.2 Semidirect products

The notion of a semidirect product of two groups generalizes the idea of the direct product. Let H and K be groups and suppose that we have an action of H on K which respects the group structure on K ; so for each $x \in H$ the mapping $u \mapsto u^x$ is an automorphism of K . Let,

$$G := \{(u, x) \mid u \in K, x \in H\}$$

and define a product on G by

$$(u, x)(v, y) := (uv^{x^{-1}}, xy)$$

for all $(u, x), (v, y) \in G$.

Note that the semidirect product of K by H , denoted by $G = K \rtimes H$ is a group under this product. It is readily seen that G contains subgroups $H^* := \{(1, x) \mid x \in H\}$ and $K^* := \{(u, 1) \mid u \in K\}$ which are isomorphic to H and K , respectively, and that $G = K^*H^*$ and $K^* \cap H^* = 1$. Moreover, K^* is normal in G and the way that H^* acts on K^* by conjugation reflects the original action of H on K , namely,

$$(1, x)^{-1}(u, 1)(1, x) = (u^x, 1)$$

for all $x \in H$ and $u \in K$.

3.3 Wreath Products

The notion of wreath products arises very naturally in the study of imprimitive groups. For example, let Σ be a partition of a set Ω into equal sized subsets. Then the group G of automorphisms of Σ consists of all $x \in \text{Sym}(\Omega)$ with the property that if $\Delta \subseteq \Omega$ then

$$\Delta \in \Sigma \Leftrightarrow \Delta^x \in \Sigma.$$

Clearly, G also acts on Σ . If we define B to be the kernel of this latter action, it is easy to see that B is isomorphic to the direct product of $|\Sigma|$ copies of $\text{Sym}(\Omega)$ where $\Delta \in \Sigma$, and that $G \cong B \rtimes \text{Sym}(\Sigma)$ where $\text{Sym}(\Sigma)$ acts on B by permuting the components of the elements of B in a natural way. This gives a fairly simple description of G . More generally, if H is an imprimitive group which has Σ as a system of blocks, then $H \leq G$, and so H will decompose, although the details of the decomposition are generally much more complicated for H than those for G . The wreath product construction which we now consider is a refinement of the construction we have just made for G .

Let G be a group, and H be a permutation group on a nonempty set Ω . The **wreath product** $G \wr H$ is the split extension of a base group G^Ω (the cartesian product of $|\Omega|$ copies of G) by H , where H acts on G^Ω by permuting the factors as it does to the elements of Ω . Identifying G^Ω with the set of functions $f : \Omega \rightarrow G$ denoted by $\text{Fun}(\Omega, G)$, we have $h^{-1}fh(\omega) = f(\omega h^{-1})$ for $h \in H, \omega \in \Omega$.

Suppose G acts on a set Δ . Then there are two natural actions of $G \wr H$:

- The *imprimitive action* on $\Delta \times \Omega$. The base group acts as the first coordinate, by the rule $(\delta, \omega)f = (\delta f(\omega), \omega)$, and H acts on the second coordinate in the usual way. As the name suggests, this action is imprimitive if $|\Delta|, |\Omega| > 1$: the partition $\Delta \times \Omega = \bigcup_{\omega \in \Omega} \Delta \times \{\omega\}$ is a system of imprimitivity. It is universal in the following sense.

Proposition 3.1. *Let C act imprimitively on Γ . Let Δ be a block of imprimitivity, Ω an index set for the parts of the corresponding system of imprimitivity, $G = C_\Delta^\Delta$ (where $C_{\{\Delta\}}$ is the setwise stabiliser of Δ), and $H = C^\Omega$. Then Γ can be identified with $\Delta \times \Omega$ in such a way that C^Γ is a subgroup of $G \wr H$ (in its imprimitive representation).*

- The *product action* on Δ^Ω , where the base group acts coordinate wise (that is if, $\phi \in \Delta^\Omega, f \in G^\Omega$ then $(\phi f)(\omega) = \phi(\omega)f(\omega)$), and H acts by permuting the coordinates $((\phi h)(\omega) = \phi(\omega h^{-1}))$ for $\phi \in \Delta^\Omega, h \in H$. This action is usually primitive:

Proposition 3.2. *Suppose H is transitive on Ω , and G is primitive but not regular on Δ . Then $G \wr H$, in its product action on Δ^Ω , is primitive.*

3.4 Affine General Linear groups

Let G be a group acting on a set Ω . If Ω is a field, then sets of permutations of certain natural types form subgroups of the symmetric group. Historically, these examples of permutation groups arose quite early in the subject; the first examples were given by Évariste Galois in 1830.

Let F be a field. Then the set of all permutations of F of the form :

$$t_{\alpha\beta} : \xi \mapsto \alpha\xi + \beta, (\alpha, \beta \in F \text{ and } \alpha \neq 0)$$

constitutes a subgroup of $\text{Sym}(F)$.

Proof. Suppose $A = \{t_{\alpha\beta} \mid t_{\alpha\beta} : \xi \mapsto \alpha\xi + \beta, \alpha, \beta \in F \text{ and } \alpha \neq 0\}$. Let $t, t' \in A$ i.e., $t(\xi) = \alpha_1\xi + \beta_1$ and $t'(\xi) = \alpha_2\xi + \beta_2$ for $\alpha_i, \beta_i \in F$. We have,

$$t \circ t' = t(t'(\xi)) = t(\alpha_2\xi + \beta_2) = \alpha_1(\alpha_2\xi + \beta_2) + \beta_1 = (\alpha_1\alpha_2)\xi + (\beta_2 + \beta_1) \in A.$$

Let $t^{-1} : \xi \mapsto \alpha_1^{-1}(\xi - \beta_1)$. Then,

$$\begin{aligned} t \circ t^{-1} &= t(\alpha_1^{-1}(\xi - \beta_1)) = \alpha_1(\alpha_1^{-1}(\xi - \beta_1)) + \beta_1 \\ &= (\alpha_1\alpha_1^{-1})(\xi - \beta_1) + \beta_1 = \xi - \beta_1 + \beta_1 = \xi. \end{aligned}$$

Similarly, for the right inverse. Thus, we see that $A \leq \text{Sym}(F)$. $\square$

The subgroup A is actually denoted by $\text{AGL}_1(F)$ and is called the 1-dimensional affine group over F . Note that the permutations are linear maps from $F \rightarrow F$. In the special case where F is a finite field of order q , $|\text{AGL}_1(F)| = q(q-1)$

We now adjoin a new element (which we shall denote by ∞) to F to obtain a set $\Omega := F \cup \infty$, and identify $\text{Sym}(F)$ with the stabiliser of ∞ in $\text{Sym}(\Omega)$. Then we can define a transitive extension G of $\text{AGL}_1(F)$ in $\text{Sym}(\Omega)$ in the sense that G is transitive on Ω and $G_\infty = \text{AGL}_1(F)$. The group G consists of all fractional linear mappings of the form

$$t_{\alpha\beta\gamma\delta} : \xi \mapsto \frac{\alpha\xi + \beta}{\gamma\xi + \delta} \text{ with } \alpha, \beta, \gamma, \delta \in F \text{ and } \alpha\delta - \beta\gamma \neq 0$$

with the convention that $t_{\alpha\beta\gamma\delta}(\infty) = \alpha\gamma^{-1}$ and $t_{\alpha\beta\gamma\delta}(-\delta\gamma^{-1}) = \infty$.

The higher dimensional affine groups are automorphism groups of affine geometries. The affine geometry $\text{AG}_d(F)$ consists of points and affine spaces constructed from the vector space F^d of row vectors of dimension d over the field F . The affine subspaces are the translates of the vector subspaces of F^d . Thus, if S is a k -dimensional subspace of F^d then

$$S + \beta := \{\alpha + \beta \mid \alpha \in S\}$$

is an affine subspace of dimension k for every $\beta \in F^d$. An automorphism of the affine space $\text{AG}_d(F)$ is a permutation of the set of points which maps each affine subspace to an affine subspace (of the same dimension). In other words, an affine automorphism is a permutation of the points that preserves, or respects, the affine geometry.

An *affine transformation* is an affine automorphism of an especially simple form. For each linear transformation $a \in \text{GL}_d(F)$ and vector $v \in F^d$ we define the affine transformation $t_{a,v} : F^d \rightarrow F^d$ by

$$t_{a,v} : u \mapsto ua + v.$$

Each of these mappings $t_{a,v}$ is an automorphism of the affine geometry $\text{AG}_d(F)$. The set $\{t_{a,v} \mid a \in \text{GL}_d(F), v \in F^d\}$ forms the *affine group* $\text{AGL}_d(F)$ of dimension $d \geq 1$ over F .

3.5 The Socle

A *minimal normal subgroup* of a nontrivial group G is a normal subgroup $K \neq 1$ of G which does not contain properly any other nontrivial normal subgroup of G . For example, a simple group has itself as its only minimal normal subgroup, while an infinite cyclic group has no minimal normal subgroup. The *socle* of a group G is the subgroup generated by the set of all minimal normal

subgroups of G ; it is denoted by $\text{soc}(G)$. By the usual convention, $\text{soc}(G) = 1$ if G has no minimal normal subgroups.

Since the set of all minimal normal subgroups of G is mapped into itself by every automorphism of G , the socle $\text{soc}(G)$ is a characteristic subgroup of G . Every nontrivial finite group has at least one minimal normal subgroup so has a nontrivial socle.

Example: The symmetric group S_4 has two nontrivial normal subgroups: A_4 and $N = \{(1234), (2143), (3412), (4321)\}$. But A_4 contains N , so N is the only minimal normal subgroup and the socle of S_4 is N .

Theorem 3.1. *Let G be a nontrivial finite group.*

1. *If K is a minimal normal subgroup of G , and L is any normal subgroup of G , then either $K \leq L$ or $\langle K, L \rangle = K \times L$.*
2. *There exists minimal normal subgroups $K_1, \dots, K_m$ of G such that $\text{soc}(G) = K_1 \times \dots \times K_m$.*
3. *Every minimal normal subgroup K of G is a direct product $K = T_1 \times \dots \times T_k$ where the T_i are simple normal subgroups of K which are conjugate under G .*
4. *If the subgroups K_i in (2) are all nonabelian, then $K_1, \dots, K_m$ are the only minimal normal subgroups of G . Similarly, if the T_i in (3) are nonabelian, then these are the only minimal normal subgroups of K .*

Proof. 1. Since $K \cap L \triangleleft G$ the minimality of K shows that either $K \leq L$ or $K \cap L = 1$. In the latter case $\langle K, L \rangle = KL = K \times L$ because both K and L are normal.

2. Because G is finite we can find a set $\mathcal{S} = \{K_1, \dots, K_m\}$ of minimal normal subgroups of G which is maximal with respect to the property that the subgroup H generated by $\mathcal{S}$ is a direct product $K_1 \times \dots \times K_m$. It remains to show that H is the socle; this will follow if we show that H contains all minimal normal subgroups of G . Let K be a minimal normal subgroup of G . Then (1) shows that either $K \leq H$ or $\langle K, H \rangle = K \times H$. The latter is impossible by the choice of $\mathcal{S}$. Hence, H contains all minimal normal subgroups of G .
3. Let T be a minimal normal subgroup of K . Then the conjugates $x^{-1}Tx$ of T under elements $x \in G$ are also minimal normal subgroups of K . Choose a set $\{T_1, \dots, T_k\}$ of these conjugates which is maximal with respect to the property that $L := \langle T_1, \dots, T_k \rangle$ is a direct product $T_1 \times \dots \times T_k$. Then using an argument analogous to that in (2) we see that L contains all the conjugates of T under G , and so $L \triangleleft G$. Since $1 \neq L \leq K$ and K is a minimal normal subgroup of G , we conclude that $K = L = T_1 \times \dots \times T_k$. Finally, for each T_i , the normal subgroups of T_i are clearly normal in K , so the minimality of T_i shows that it must be a simple group.
4. Suppose that G has a minimal normal subgroup K which is distinct from the K_i , ($i = 1, \dots, m$). Then (1) shows that K centralizes each of the K_i and so $K \leq Z(\text{soc}(G))$. However, if each K_i is nonabelian, then $Z(K_i) = 1$ by (3). This implies that $Z(\text{soc}(G)) = 1$ and so $K = 1$ contrary to the choice of K .

□

3.6 Classes of finite primitive permutation groups

Before stating our theorem we describe various classes of primitive permutation groups as described in [5]. In what follows, X will be a primitive permutation group on a finite set Ω of size n , and α a point in Ω . Let B be the socle of X , Then $B \cong T^k$, $k \geq 1$, where T is a simple group. We denote by $A \cdot B$ an extension (not necessarily split) of A by B .

I. *Affine Groups*. Here $T = Z_p$ for some prime p , and B is the unique minimal normal subgroup of X and is regular on Ω of degree $n = p^k$. The set Ω can be identified with $B = Z_p^k$ so that X is a subgroup of the affine group $AGL(k, p)$ with B the translation group and $X_\alpha = X \cap GL(k, p)$ irreducible on B .

II. *Almost simple groups*. Here $k = 1$, T is a nonabelian simple group and $T \leq X \leq \text{Aut}(T)$. Also $T_\alpha \neq 1$.

III. In this case $B \cong T^k$, $k \geq 2$ and T a nonabelian simple group. We distinguish three types:

III(a). *Simple Diagonal action*. Define

$$W = \{(a_1, \dots, a_k) \mid a_i \in \text{Aut}(T), \pi \in S_k, a_i \equiv a_j \pmod{\text{Inn } T} \text{ for all } i, j\}$$

where $\pi \in S_k$ just permutes the components a_i naturally. With the obvious multiplication, W is a group with socle $B \cong T^k$, and $W = B \cdot (\text{Out}T \times S_k)$, a (not necessarily split) extension of B by $\text{Out}T \times S_k$. We define an action of W on Ω by setting

$$W_\alpha = \{(a, \dots, a) \cdot \pi \mid a \in \text{Aut}T, \pi \in S_k\}.$$

Thus $W_\alpha \cong \text{Aut}T \times S_k$, $B_\alpha \cong T$ and $n = |T|^{k-1}$.

For $1 \leq i \leq k$ let T_i be the subgroup of B consisting of the k -tuples with 1 in all but the i -th component, so that $T_i \cong T$ and $B = T_1 \times \dots \times T_k$. Put $\tau = \{T_1, \dots, T_k\}$, so that W acts on τ . We say that the subgroup X of W is of type III(a) if $B \leq X$, and letting P be the permutation group X^τ , one of the following holds:

1. P is primitive on τ .
2. $k = 2$ and $P = 1$.

We have $X_\alpha \lesssim \text{Aut}T \times P$, and $X \leq B \cdot (\text{Out}T \times P)$. Moreover, in case (1), B is the unique minimal normal subgroup of X , and in case (2), X has two minimal normal subgroups T_1 and T_2 , both regular on Ω .

III(b). *Product action*. Let H be a primitive permutation group on a set Γ , of type II or III(a). For $l > 1$, let $W = H \wr S_l$, and take W to act on $\Omega = \Gamma^l$ in its natural product action. Then for $\gamma \in \Gamma$ and $a = (\gamma, \dots, \gamma) \in \Omega$ we have $W_\alpha = H_\gamma \wr S_l$, and $n = |\Gamma|^l$. If K is the socle of H then the socle B of W is K^l , and $B_\alpha = (K_\gamma)^l \neq 1$.

Now W acts naturally on the l factors in K^l , and we say that the subgroup X of W is of type III(b) if $B \leq X$ and X acts transitively on these l factors.

Finally, one of the following holds:

1. H is of type II, $K \cong T$, $k = l$ and B is the unique minimal normal subgroup of X ,

2. H is of type III(a), $K \cong T^{k/l}$ and X and H both have m minimal normal subgroups, where $m \leq 2$; if $m = 2$ then each of the two minimal normal subgroups of X is regular on Ω .

III(c). *Twisted wreath action.* Here X is a twisted wreath product $T \wr_\phi P$ defined as follows. Let P be a transitive permutation group on $\{1, \dots, k\}$ and let Q be the stabiliser P_1 . We suppose that there is a homomorphism $\phi : Q \rightarrow \text{Aut} T$ such that $\text{Im} \phi$ contains $\text{Inn} T$. Define

$$B = \{f : P \rightarrow T \mid f(pq) = f(p)^{\phi(q)} \text{ for all } p \in P, q \in Q\}.$$

Then B is a group under pointwise multiplication, and $B \cong T^k$. Let P act on B by

$$f^p(x) = f(px) \text{ for } p, x \in P.$$

We define $X = T \wr_\phi P$ to be the semidirect product of B by P with this action, and define an action of X on Ω by setting $X_\alpha = P$. We then have $n = |T|^k$, and B is the unique minimal normal subgroup of X and acts regularly on Ω .

We say that the group X is of type III(c) if it is primitive on Ω .

- REMARKS. 1. The classes I, $\dots$, III(c) are pairwise disjoint; this is clear from the differing structures and actions of the socles B on Ω .
2. Although III(b) is the only case where X is described as a subgroup of a group with a product action, some of the groups of other types are subgroups of a wreath product $S_a \wr S_b$ with product action on Ω (and $n = a^b, a > 1, b > 1$); these are:
 1. groups of type I where X_α is imprimitive as a linear group, b divides k and $a = p^{k/b}$,
 2. groups of type III(c), with $a = |T|$ and $b = k$; in this case X is contained in the wreath product $H \wr S_k$, where $H = T \times T$ is of type III(a); note that here the socle of $H \wr S_k$ is isomorphic to $B \times B$.
 3. Although groups of type III(c) are the only groups describes as twisted wreath product, various primitive groups of types I, III(a) or III(b) may also be nontrivial twisted wreath products as abstract groups. The distinguishing feature of III(c) is the existence of unique nonabelian regular normal subgroup.

Theorem 3.2. *Any finite primitive permutation group is permutation equivalent to one of the types I, II, III(a), III(b) or III(c) described in 3.6.*

The next section sketches a few of the results obtained by using 3.2 including the Sim's Conjecture.

4 Application of the O’Nan Scott Theorem

With the O’Nan-Scott Theorem available, a problem about finite primitive group G can be broken up into these five cases. In a typical situation, we can deal with the case of a regular normal subgroup in a straightforward way. If G is of diagonal type we have a detailed description of the action, while if G is of product type a strong inductive setup is available.

So often a problem can be reduced to the case of a group G of almost simple type. At this point we turn to the large body of detailed knowledge available about finite simple groups, we can consider the separate types of finite simple group as possible socles for G . Of course, in a particular problem any or all of these steps may be nontrivial, but the O’Nan-Scott Theorem does provide an effective framework for using detailed information about finite simple groups to answer significant questions about finite primitive groups.

4.1 Listing Primitive Groups

The analysis of primitive groups in terms of their socles provides a natural approach to listing the primitive groups. These lists are constructed in the following way. If G is primitive on Ω then the socle H is transitive and $H \triangleleft G \leq \text{Sym}(\Omega)$. Each list item is essentially a transitive action for a particular socle H along with information about the structure of G_α and the normalizer of H in $\text{Sym}(\Omega)$. Such a list item is called a *cohort* of groups. The permutation groups in one cohort all have the same socle with a specified action. There are 762 cohorts of proper primitive groups of degree less than 1000.

A primitive group G has a socle $H = T^m$ for some simple group T . If H is abelian then G_α is an irreducible subgroup of $GL_m(p)$ and we do not explore this case any further. In the case of a nonabelian regular socle, the degree of G is $|T|^m$ where $m \geq 6$. Since the order of a nonabelian simple group is even and at least 60, the degree of G is even and at least $60^6 > 1000$. Thus neither of the lists includes any primitive groups of this type. The degree of a group G of diagonal type is also a power $|T|^{m-1}$ of the order of a nonabelian simple group and hence is even. The primitive groups of this type with degrees less than 1000 have $m = 2$ and $T = A_5, A_6, PSL_2(7), PSL_2(8)$, or $PSL_2(11)$.

If G is of product type then the degree of G is a power d^k , with $k \geq 2$, where d is the degree of some other primitive group U . The group G has odd degree when U has odd degree. The condition $d^k < 1000$ requires $k = 3$ and $d < 10$, or $k = 2$ and $d < 32$. Thus the groups of product type give an inductive class of examples in each list. There are 74 cohorts of groups with socles of this type and degree less than 1000.

The largest class of examples on both lists consists of the groups with a simple nonabelian socle. For these groups, we need information about the maximal subgroups of the almost simple groups. Concerted efforts by a number of mathematicians over the past century have provided enough information to deal with this case.

4.2 Degree and Rank

The theorem 3.2 shows that a primitive group which is not of almost simple type has degree which is a power of the order of a finite simple group or a nontrivial power d^k of the degree d of some other primitive group. Thus the degrees that actually occur for proper primitive groups are a relatively sparse set of natural numbers. There are 486 degrees $n < 1000$ such that the only primitive group

of degree n . Then E is the union of the following sets:

$$E_1 := \{p \mid p \text{ prime}, p \geq 5\};$$

$$E_2 := \{m^k \mid m \geq 2, k \geq 2, m^k \geq 4\};$$

$$E_3 := \{n \mid \text{there is nonabelian simple group of order } n\};$$

$$E_4 := \{d \mid \text{there is a proper primitive group } G \text{ of degree } d \text{ whose socle is simple and nonabelian}\}.$$

The density of each of these sets can be estimated. The sets E_1 and E_2 involve only properties of the integers and we only need the orders of the finite simple groups to deal with set E_3 . On the other hand, many detailed facts about finite simple groups are required to estimate the density of the set E_4 . Calculating these densities calculated by Cameron et al. [6] obtained the following asymptotic estimate for the density of E . Let $\pi(x)$ denote the number of primes $p \leq x$.

Theorem 4.1. *If $e(x)$ is the number of degrees $n \leq x$ such that there is a proper primitive group of degree n then*

$$e(x) = 2\pi(x) + (1 + \sqrt{2})x^{\frac{1}{2}} + O\left(\frac{x^{\frac{1}{2}}}{\log x}\right) \sim \frac{2x}{\log x}$$

As another example of the reduction problem to the almost simple case, consider the question of rank. Suppose that G is a primitive permutation group with a nonabelian socle $H = T^m$ with T simple. Then the rank r of G satisfies $r \geq m + 1$. This follows from the fact that if G is a primitive subgroup of the wreath product $U \wr S_b$ where U is a primitive group with rank r_0 , then rank of G is at least $\binom{r_0 + b - 1}{b} \geq b + 1$ and if G is primitive with a nonabelian regular socle $H = T^m$ then, G has rank at least $m + 1$.

When studying a 2-transitive group G , these lower bounds on the rank shows that either G is almost simple, or else G is an affine group. Then analysis shifts, on the one hand, to examining the 2-transitive actions of the almost simple groups and, on the other hand, to determining the subgroups of $GL_m(p)$ that act transitively on the set of nonzero vectors in the underlying vector space. In this way, the finite 2-transitive groups have been completely determined.

4.3 The Sims Conjecture

Suppose that G is a primitive group of degree n acting on set Ω . The subdegrees of G are the lengths of the orbits of the stabilizer G_α . Consider the primitive groups with a given subdegree $d > 1$. For a fixed d there is no immediate bound on the degree n of G . For example, for any prime p the dihedral group D_{2p} has a representation as a primitive permutation group of degree p with a subdegree $d = 2$. On the other hand, a small subdegree does strongly restrict the structure of G_α . Following this investigation of the cases $d = 3$ and $d = 4$, C.Sims([3]) was lead to conjecture the following theorem.

Theorem 4.2. *There is a function $f : \mathbb{N} \mapsto \mathbb{N}$ such that if G is a finite primitive group with a suborbit of length $d > 1$ then the pointwise stabilizers have order at most $f(d)$.*

The simplest case in proving the Sims Conjecture is when G is a primitive group with a regular socle H . We can identify Ω with the elements of H in such a way that G_α acts on H by conjugation in the same way it acts on Ω . Since G is primitive there are no nontrivial G_α -invariant subgroups of H . Thus the group generated by elements of H in any nontrivial orbit of G_α is H itself. In

particular, G_α acts faithfully on each of its orbits and hence has order at most $d!$. This establishes the result in the case of a regular socle.

If G is of diagonal type then an analysis of the action itself, without using specific properties of the simple group T , shows that $|G_\alpha| \leq (d!)^{d+1}$ in this case. If G is a primitive group of product type then G is permutation isomorphic to a subgroup of a wreath product of the form $U \wr S_b$ where U is primitive on a set Δ and is of almost simple or diagonal type. Suppose $\delta \in \Delta$. Then it can be shown that $|G_\alpha| \leq |U_\delta|^d d!$. Thus if we assume that there is an increasing function $h(d)$ making 4.2 true for all groups G with a simple socle then we can define a function $f(d)$ that will work for all primitive groups. The proof of 4.2 is then completed by dealing with the almost simple case and using key results about the subgroup structure of G_α . This is the most complicated part of the proof and uses specific information about various simple groups. The function $f(d)$ can be taken of the form $\exp(d^2 o(d))$ though this is not best possible.

5 Conclusion

It is unfortunate that we had to omit the study of the generalisation to the Sim's Conjecture which was proposed by Pablo Spiga in [7], however, the theory of permutation groups and classification of finite primitive permutation group is a fascinating area of research on its own due to the immense applicability of these groups not only in mathematics but also in physics. There are many more applications of the O'Nan-Scott Theorem like studying combinatorial objects such as graphs, random generation of alternating groups etc.

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